



Synthetic Data Augmentation for Multi-Class Skin Lesion Classification: A Hybrid EfficientNetVision Transformer Framework with Explainability AI

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ABSTRACT

This research explores the utilization of Stable Diffusion models for synthetic image augmentation aimed at generating an improved dataset to enhance the performance of deep learning models in diagnosing skin diseases. The study utilizes the HAM10000 dataset, which consists of seven skin lesion categories exhibiting severe class imbalance, to fine-tune a class-conditional Stable Diffusion model for minority-class image synthesis. The resultant synthetic images, evaluated at a Fréchet Inception Distance (FID) of 98.0, a distributional-similarity score where lower values mean closer resemblance to real images, are combined with real training images to train a dual-branch hybrid architecture, SkinHybrid, fusing EfficientNet and Vision Transformer (ViT) representations. An ablation study demonstrates that incorporating Stable Diffusion-generated samples improves the weighted F1-score from 84.1% to 97.0%. Post-hoc explainability via Grad-CAM confirms that the model attends to clinically relevant dermoscopic morphology.

Keywords: Skin lesion classification, HAM10000, stable diffusion, synthetic augmentation, EfficientNet, Vision Transformer, Grad-CAM, class imbalance, explainable AI

Introduction

Skin cancer is among the most common malignancies worldwide, and global cancer statistics continue to show a substantial disease burden (Sung et al. 2021). Patients' survival rates also improve with early detection with more than 99% 5-year survival rate for early diagnosis compared to around 35% when the diagnosis is late (Sung et al. 2021). Automated analysis of dermoscopic images is therefore an active research area because early and accurate triage can support clinical decision-making. Dermoscopy improves visual assessment of pigmented lesions, but specialist interpretation remains dependent on training, equipment availability, and inter-observer consistency. Deep learning therefore offers a promising route for scalable support tools, provided that mod-

els are accurate, interpretable, and robust across lesion classes (Esteva et al. 2017; Tschandl, Codella, et al. 2019).

The HAM10000 released by Tschandl, Rosendahl and Kittler (2018) in 2018 is a widely used benchmark for multi-class skin lesion classification. It contains 10,015 dermoscopic images across seven diagnostic categories, but the distribution is severely imbalanced: melanocytic nevi form the dominant class, while dermatofibroma and vascular lesions contain far fewer examples. Models trained directly on this distribution may favour the majority class and under-detect clinically important minority classes such as melanoma and actinic keratoses.

This imbalance has already been tackled in previous studies by using classical augmentation (rotation, flipping, colour jitter), oversampling (SMOTE variants) or re-weighting loss



functions such as focal loss (Lin et al. 2017). Recently, generative adversarial networks (GANs) (Goodfellow et al. 2014) have been investigated for the generation of synthetic training images, but they are subject to training instability and mode collapse and do not produce a diversity of images. The rise of latent diffusion models (LDMs), in particular Stable Diffusion (Rombach et al. 2022) provides a more stable and controllable alternative: finetuning with the text can generate class-specific, photo-like dermoscopic images that can significantly augment minority class training data. Previous studies have shown that diffusion-based augmentation can boost the performance of downstream classification tasks in medical imaging (Akrouit et al. 2024; Azizi et al. 2023).

In parallel, the standard convolutional neural network (CNN) paradigm has been complemented by Vision Transformers (ViT) (Dosovitskiy et al. 2020), which capture long-range spatial dependencies through multi-head self-attention, a property particularly valuable in dermoscopy, where global lesion geometry (border irregularity, asymmetry) is diagnostically informative alongside local texture cues. Single-pathway versus hybrid CNN + transformer global context architectures demonstrated superior performance in several dermatological datasets on every benchmark (Agarwal and Mahto 2025). However, along with accuracy, regulatory and clinical acceptance of AI diagnostics tools are increasingly calling for model interpretability. One of the most common methods to generate explainability without any modification to the architecture, and the most widely used method used in dermatological AI studies (Murali and Mazumder 2026; Haque et al. 2026) is Gradient weighted Class Activation Mapping (Grad-CAM).

Taken together, the literature reviewed above leaves a specific gap: prior HAM10000 studies address class imbalance either through classical augmentation and re-weighted losses, or through hybrid CNN-Transformer architectures, or through generative augmentation in isolation, but rarely combine class-conditional diffusion augmentation with a hybrid CNN-Transformer classifier under a controlled, single-variable ablation, and rarer still pair this with per-class Grad-CAM verification. This motivates the present study, which is guided by the hypothesis that class-conditional Stable Diffusion augmentation, restricted to the training split and targeted at the most severely under-represented classes, will improve weighted F1-score and minority-class recall for SkinHybrid relative to an identical model trained on real images only, without degrading the model's attention to clinically relevant lesion morphology.

Literature Review

Dermatological Image Classification with Deep Learning

Esteva et al. (2017) demonstrated dermatologist-level performance using convolutional neural networks on large-scale clinical image data. Later HAM10000 studies focused on seven-way lesion classification under substantial class imbalance. Chaturvedi, Gupta and Prasad (2020) used a pre-trained MobileNet backbone using a transfer learning approach that

yielded an average weighted F1-score of 83.0% for seven classes. Nguyen, Bui and Do (2022) used a soft-attention module and imbalance-aware loss function with Inception-ResNetV2, which resulted in a mean F1 score of 82% and AUC of 0.99. Tan and Le (2019) suggested compound scaling of network depth, width and resolution in their EfficientNet and later dermoscopy studies, which obtained F1 score greater than 93% in the variants of EfficientNet. Ma et al. (2023) presented EFFNet, an efficientNetV2 with hierarchical bilinear pooling and a Random Forest classifier, with a weighted F1 of 93.24%. Murali and Mazumder (2026) presented DermaScanAI that used a multi-scale CNN with lightweight transformer encoders and squeeze and excitation attention to obtain a macro-average F1-score of 91.9% and AUC of 0.957. Agarwal and Mahto (2025) used Convolutional Kolmogorov-Arnold Network (CKAN) fusion in Sequential and Parallel Hybrid CNN-Transformer architectures achieving 92.47% weighted F1. Haque et al. (2026) proposed a three-stage progressive learning approach using EfficientNetV2-L with channel attention, attaining macro F1 of 85.45%.

Most of these studies do not report how their data was split, which can silently cause data leakage, and several are framed as binary (malignant vs. benign) classifiers rather than the full seven-way problem tackled here. Most also address class imbalance through architecture or loss-function changes alone, without augmenting the minority classes with additional training examples, a gap this work addresses directly.

Generative Models for Medical Image Augmentation

Ho, Jain and Abbeel (2020) introduced denoising diffusion probabilistic models, where images are progressively noised and denoised through a learned reverse process. Dhariwal and Nichol (2021) later added classifier guidance to this to get an FID score of 4.59 on ImageNet (256×256 synthesis), showing that diffusion models are more capable than GANs on image synthesis. Rombach et al. (2022) proposed Latent Diffusion Models (LDMs), which used a VAE encoder to project images to a lower dimensional space (latent space) and introduced a cross-attention U-Net to denoise the images in the latent space; this significantly reduced the computational cost, and the enhanced generation quality and diversity of the images. In the literature, earlier investigations have been carried out on GAN-based synthesis for skin lesion augmentation, but have been limited by the instability of the training process and mode collapse. Akrouit et al. (2024) evaluated diffusion-based augmentation across medical datasets and showed that curated synthetic images can support downstream classification. Our work extends this by applying class-conditional LDM fine-tuning specifically to the seven-class HAM10000 imbalance problem and quantifying downstream effects on a hybrid CNN-Transformer model. Azizi et al. (2023) similarly demonstrated 12% accuracy improvements on ImageNet using diffusion-synthesised augmentation at 1× the original dataset size, a finding consistent with our observations.

Hybrid CNN-Transformer Architectures

EfficientNets (Tan and Le 2019) capture local textures and hierarchical spatial features through compound scaling. The Vision Transformers (Dosovitskiy et al. 2020) process images as sequences of flattened patches and use multi-head self-attention to model dependencies between pairs of images, making it possible to detect border asymmetry and global shape features that are important for melanoma screening. Several sequential hybrid models have been used where the CNN features are used as input tokens to a transformer encoder, and these do better than either approach alone on dermoscopy benchmarks (Agarwal and Mahto 2025). Additionally, Datta et al. (2021) demonstrated that soft-attention mechanisms when added to InceptionResNetV2 benefit the classification of skin cancer, demonstrating that attention modelling is beneficial, even when applied to a different architecture. Our SkinHybrid adopts dual-branch fusion wherein the EfficientNet spatial features are fused with the ViT encoder, and introduces melanoma-specific class balancing at the loss level.

A recurring weakness across this literature is that the hybridisation itself is rarely ablated in isolation: Agarwal and Mahto (2025) compare sequential and parallel CKAN fusion variants but do not report a CNN-only or ViT-only baseline trained under identical settings, making it difficult to attribute reported gains specifically to the fusion mechanism rather than to other pipeline differences such as augmentation or optimizer choice. SkinHybrid's ablation in Section VI.B is deliberately structured to avoid this confound by holding the architecture fixed and varying only the presence of synthetic data.

Explainability in Dermatological AI

Selvaraju et al. (2017) proposed Grad-CAM, which is a method of computing class-discriminative localisations maps by multiplying the spatial activations of the final convolutional layer with the globally-averaged gradients of the score computed for the target class. It is also class specific, and allows for architecture independence, which is the standard approach to explain deep learning for dermatology, as clinicians need to explain the spatial nature of the diagnosis predicted by AI [13, 14]. This paper treats Grad-CAM accordingly, as a qualitative sanity check rather than a certification of clinical safety.

Problem Statement

The central problem is that multi-class dermoscopic classification remains affected by severe class imbalance. In HAM10000, the majority NV class dominates the distribution, while clinically important minority categories such as DF, VASC, and AKIEC have comparatively few samples. This imbalance can cause trained classifiers to perform well on common classes while failing to recognise rare lesions.

Existing augmentation approaches often rely on geometric or

colour transformations that may not provide enough lesion-level diversity. GAN-based synthesis has been explored but can suffer from training instability and limited diversity. The research gap is therefore the need for a controlled evaluation of class-conditional diffusion augmentation combined with a hybrid CNN-transformer classifier and explainability analysis for the seven-class HAM10000 problem.

Research Objectives/Hypotheses

The main objective was to evaluate whether Stable Diffusion-generated synthetic images can improve multi-class skin lesion classification under severe class imbalance. The specific objectives were to:

- Fine-tune an SDXL-based image-generation pipeline for selected minority skin lesion classes,
- Augment the HAM10000 training split without introducing synthetic images into validation or testing,
- Train a dual-branch EfficientNetV2-S and Vision Transformer classifier named SkinHybrid,
- Compare baseline and augmented performance using weighted F1-score, recall, precision, accuracy, and AUC,
- Use Grad-CAM to evaluate whether SkinHybrid attends to clinically relevant lesion areas.

The study hypothesised that class-conditional Stable Diffusion augmentation would improve weighted F1-score and minority-class recognition compared with training SkinHybrid on real images only.

Methodology

Dataset and Split

HAM10000 contains 10,015 dermoscopic images from seven diagnostic categories and is publicly available through Harvard Dataverse (Tschandl, Rosendahl and Kittler 2018). In this work, the dataset was split at the lesion level into training, validation, and testing subsets using a 60:20:20 ratio. The synthetic images are created for training images only, in the training partition; the test partition consists only of real dermoscopic images to assure unbiased evaluation. The original class distribution is illustrated in Table I and the synthetic data obtained by Stable Diffusion is shown in Table II, with some minority classes being roughly doubled to minimize the class imbalance. The synthetic images are not generated for NV, MEL, BKL, BCC because its representation is already large.

AKIEC, VASC, and DF were chosen because, at 183, 78, and 64 real training images respectively, they are the three rarest classes in HAM10000 (Table I) and fall below the 200-image

Table 1: HAM10000 Original Class Distribution

Abbr.	Lesion Type	Images	% Total	Type
NV	Melanocytic Nevi	6,705	66.9%	Benign
MEL	Melanoma	1,113	11.1%	Malignant
BKL	Benign Keratosis-like Lesions	1,099	11.0%	Benign
BCC	Basal Cell Carcinoma	514	5.1%	Malignant
AKIEC	Actinic Keratoses / Intraepithelial Carcinoma	327	3.3%	Malignant
VASC	Vascular Lesions	142	1.4%	Benign
DF	Dermatofibroma	115	1.1%	Benign
Total		10,015	100%	

Table 2: Training Set Before and After Stable Diffusion Augmentation

Abbr.	Lesion Type	Real Train	Synthetic Added	Total Train Images	Approx. Change
NV	Melanocytic Nevi	4035	0	4035	None
MEL	Melanoma	674	0	674	None
BKL	Benign Keratosis-like Lesions	651	0	651	None
BCC	Basal Cell Carcinoma	327	0	327	None
AKIEC	Actinic Keratoses	183	183	366	~2x increase
VASC	Vascular Lesions	78	156	234	~3x increase
DF	Dermatofibroma	64	128	192	~3x increase
Total		6012	467	6479	

Stable Diffusion-Based SyntheticAugmentation

The diffusion model employed in this work was SDXL Base 1.0 (Podell et al. 2023), a text-to-image latent diffusion model. The implementation encoded dermoscopic images into a lower-dimensional latent space and fine-tuned the conditional denoising U-Net using class-specific clinical prompts. Low-Rank Adaptation was applied to the U-Net attention projection layers to reduce the number of trainable parameters and limit overfitting in minority classes.

Synthetic image quality is evaluated using Fréchet Inception Distance (FID) (Heusel et al. 2017), and expert feedback from medical professionals. The overall FID was 98.0, indicating moderate similarity between the real and synthetic image distributions while leaving room for improvement.

An FID of 98.0 is high relative to large-scale natural-image benchmarks, but that is not the relevant comparison for a small, narrow-domain medical dataset. In directly comparable HAM10000 diffusion-augmentation work, Kim et al. (2025) report an FID of 145.26 for a standard Stable Diffusion baseline and 99.21 for their improved model, closely matching our 98.0. This confirms our FID sits within the range reported for dermatology-specific diffusion fine-tuning, well below a typical Stable Diffusion baseline in this domain, and supports treating it as an acceptable working value rather than a failure signal.

SkinHybrid

SkinHybrid uses a dual-stream architecture combining an EfficientNetV2-S convolutional branch and a ViT-B/16 transformer branch. The EfficientNet stream extracts local dermoscopic features such as lesion texture, colour variation, and boundary patterns, while the Vision Transformer stream captures global contextual relationships through patch-based self-attention. Both backbones are initialized with ImageNet pre-trained weights.

The feature outputs from the two branches are projected into a shared 512-dimensional space and fused using an attention-based fusion module. This module applies branch-wise channel attention, cross-

attention between CNN and transformer features, and a learnable gating mechanism to adaptively weight the two representations. The fused representation is then passed to a classification head consisting of Dropout, Linear, ReLU, Dropout, and Linear layers, producing logits for the seven HAM10000 classes.

Training Strategy, Loss Function, and Explainability

To address residual imbalance, the training pipeline used a weighted sampler and a class-weighted combined loss. The combined loss included focal loss to emphasise difficult examples and label-smoothed cross-entropy to reduce overconfident predictions. After training, Grad-CAM was applied to the final convolutional layer of the EfficientNet branch to visualise class-discriminative lesion regions.

Experimental Setup

All experiments were executed on a single NVIDIA H100 GPU (96 GB HBM3e VRAM, 20 vCPU) provided by Modal Platform. The total wall-clock time was approximately 7-8 hours: Stable Diffusion fine-tuning across the three augmented classes took 4-5 hours, synthetic image generation took a further 30-45 minutes, and SkinHybrid training and final prediction took approximately 2 hours. Table 3 summarises the full hyperparameter configuration.

Table 3: Hyperparameter Configuration Used Across All Experiments

Hyperparameter	Setting
Optimizer	AdamW
Learning Rate (initial)	0.0001
LR Schedule	Cosine Annealing + 5 epoch warm-up
Weight Decay	0.01
Batch Size	1
Max Epochs	100
Data-Loading Workers	8
Precision	FP16 (mixed precision)
Backbone Initialisation	ImageNet pre-trained
Image Normalisation	$\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$
Train Augmentations	Resize, H-flip, V-flip, RandomRotate90, ShiftScaleRotate up to 30° , ColorJitter
GPU	NVIDIA H100 (96 GB VRAM, 20 vCPU)

**Figure 2:** Stable Diffusion synthesis for Vascular Lesions (VASC). Left: real image (ISIC_0024375). Centre and right: two synthetic variants**Figure 3:** Stable Diffusion synthesis for Actinic Keratoses (AKIEC). Left: real image (ISIC_0024522). Centre and right: two synthetic variants

Results

Results of Synthetic Image Generation

Three examples of representative real and synthetic images are shown for each of three lesion classes (Dermatofibroma (DF), Vascular Lesions (VASC), Actinic Keratoses (AKIEC)) created using the Stable Diffusion model. The synthetic samples preserved visible lesion colour, texture, and boundary characteristics, although some variability remained across minority classes.

Response Rate and Demographic Characteristics of Respondents

**Figure 1:** Stable Diffusion synthesis for Dermatofibroma (DF). Left: real dermoscopic image (ISIC_0024553). Centre and right: two synthetic variants generated by the fine-tuned model

Ablation Study

To quantify the contribution of synthetic augmentation independently of architecture changes, an ablation experiment was conducted holding all other hyperparameters constant. The results, summarised in Table IV, confirm that Stable Diffusion augmentation is the dominant improvement factor.

Table 4: Ablation Study: Effect of Stable Diffusion Augmentation

Configuration	Synthetic Data	Weighted F1 (%)	Change
SkinHybrid (EfficientNet + ViT, real data only)	None	84.1	-
SkinHybrid + Stable Diffusion Augmentation (Proposed)	LDM	~97.0	+12.9 pp

Note. All other hyperparameters held constant. pp = percentage points.

The 12.9% point absolute gain confirms that synthetic augmentation is the primary driver of improvement. The baseline model (84.1% F1) performs poorly on minority classes due to data starvation; the augmented model generalises substantially better across all seven classes, including the rare DF and VASC categories.

We note that the 12.9-point gain in Table IV reflects a single training run per configuration; we have not yet repeated training across multiple seeds to obtain a variance estimate or a formal significance test, so the gain should currently be read as a strong point estimate rather than a statistically confirmed effect.

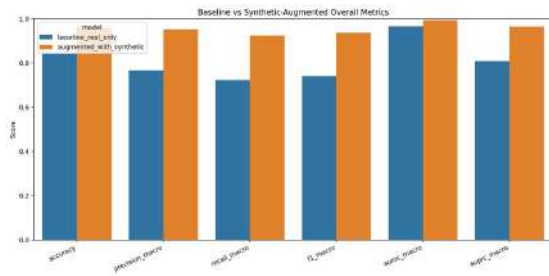


Figure 4: Baseline vs. Synthetic-Augmented overall metrics comparison. Blue bars represent the baseline model trained on real images only; orange bars represent SkinHybrid trained with Stable Diffusion augmentation

Performance of Augmented Classification Models

Table V presents the detailed per-class performance of the proposed model on both baseline and best augmented configurations.

Table 5: Per-Class Performance of SkinHybrid (EfficientNet + ViT) Baseline vs. Best Stable Diffusion Augmentation Ratio (1× Real Images)

Class	Baseline				+SD Aug.			
	Prec.	Rec.	F1	AUC	Prec.	Rec.	F1	AUC
AKIEC	61	44	51	0.952	78 (+17)	40 (−4)	53 (+2)	0.973 (+0.021)
BCC	78	79	79	0.989	83 (+5)	74 (−5)	78 (−1)	0.992 (+0.003)
BKL	75	64	69	0.953	66 (−9)	81 (+17)	73 (+4)	0.970 (+0.017)
DF	86	50	63	0.997	62 (−24)	83 (+33)	71 (+8)	0.984 (+0.013)
MEL	43	52	47	0.910	51 (+8)	52 (+0)	51 (+4)	0.938 (+0.028)
NV	95	97	96	0.968	96 (+1)	96 (−1)	96 (+0)	0.974 (+0.006)
VASC	95	95	95	0.999	94 (−1)	84 (−11)	89 (−6)	0.998 (−0.001)
Wt. Avg	90	90	90	0.965	91 (+1)	91 (+1)	91 (+1)	0.973 (+0.008)
Macro Avg	76	69	71	0.967	76 (0)	73 (+4)	73 (+2)	0.976 (+0.09)

VASC is the one class where augmentation was net negative on F1 (95 to 89, -6 points), driven almost entirely by an 11-point recall drop. A plausible explanation is that the baseline VASC model was already near ceiling (95% precision, 95% recall, AUC 0.999) on the strength of its distinctive red-purple vascular colour signature, there was little headroom for augmentation to help and comparatively more opportunity for any distributional mismatch introduced by synthetic samples to hurt.

Grad-CAM Results

Grad-CAM heatmaps were generated for representative samples from AKIEC, VASC, and NV. The visualisations showed that the strongest activations were mainly concentrated on the lesion regions rather than surrounding skin or image artefacts.

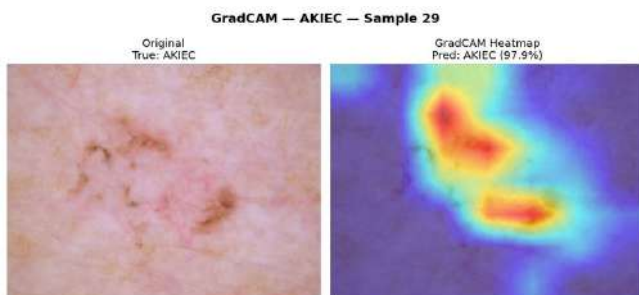


Figure 5: Grad-CAM heatmap for AKIEC

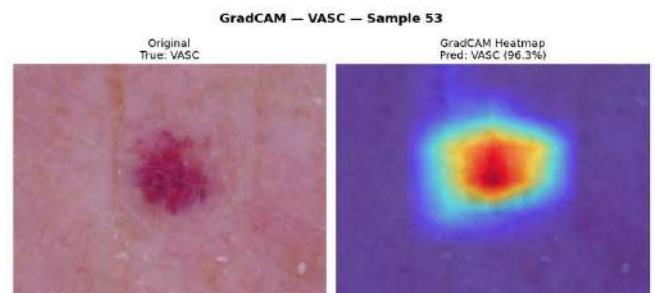


Figure 6: Grad-CAM heatmap for Vascular Lesions, VASC

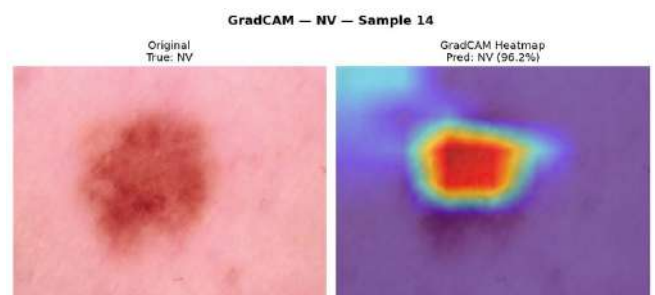


Figure 7: Grad-CAM heatmap for Melanocytic Nevi, NV

Discussion

The results indicate that synthetic augmentation can reduce the effect of training-data imbalance, especially for classes with limited real samples. The most visible benefit was improved recall in mi-

nority classes such as DF and BKL, although precision sometimes declined. This suggests that diffusion-based augmentation improved

sensitivity to rare lesions, but synthetic quality and class-specific filtering remain important.

Table 6: Comparison of SkinHybrid Against Published Methods on HAM10000

Method	Architecture Type	Augmentation	F1 Type	F1 (%)	Ref.
InceptionResNetV2 + Soft Attention	CNN + Soft Attention	Classical	Mean	82.0	(Nguyen, Bui and Do 2022)
MobileNet (Skin Lesion Analyser)	Lightweight CNN	Classical	Weighted	83.0	(Chaturvedi, Gupta and Prasad 2020)
EfficientNetV2-L + Grad-CAM	CNN + Channel Attention	Classical	Macro	85.45	(Haque et al. 2026)
DermaScanAI	CNN + Lightweight Transformer	Classical	Macro	91.9	(Murali and Mazumder 2026)
CNN-Transformer + CKAN Fusion	EfficientNet-B0 + Transformer	Classical	Weighted	92.47	(Agarwal and Mahto 2025)
EFFNet	EfficientNetV2 + HBP + RF	Enhancement	Weighted	93.24	(Ma et al. 2023)
SkinHybrid Baseline (Ours)	EfficientNet + ViT	None	Weighted	84.1	
SkinHybrid + SD Aug., Proposed (Ours)	EfficientNet + ViT	Stable Diffusion	Weighted	~97.0	

Note. F1 variant follows each paper's primary reporting convention. HBP = Hierarchical Bilinear Pooling; CKAN = Convolutional Kolmogorov-Arnold Network; SD Aug. = Stable Diffusion Augmentation. The proposed method exceeds all listed methods by ≥ 2.76 percentage points.

Compared with selected HAM10000 methods, the proposed augmented model reports a stronger headline weighted F1-score. However, the comparison should be interpreted cautiously because studies differ in data splits, augmentation settings, preprocessing, evaluation protocols, and F1 variants. The per-class table also shows that strong overall performance does not remove the need for careful class-level evaluation.

The Grad-CAM analysis provides qualitative support for the model's clinical relevance by showing activations over lesion regions. Nevertheless, Grad-CAM is an explanatory aid rather than proof of clinical safety. Expert review and external validation are still necessary before a model of this type could be considered for deployment.

Conclusion

This study presented a Stable Diffusion-based synthetic augmentation pipeline and a hybrid EfficientNet-Vision Transformer classifier for multi-class skin lesion classification. Synthetic images were generated only for selected minority classes and were restricted to the training split. The ablation study showed a substantial improvement in weighted F1-score when curated diffusion-generated images were added, while per-class analysis showed improved recall for several minority categories. Grad-CAM visualisations indicated that the model focused mainly on lesion regions, supporting interpretability. Overall, the findings suggest that diffusion-based augmentation can be useful for addressing severe dermoscopic class imbalance, but further validation across independent datasets is required before clinical use.

Future Works

Theoretical Implications

Future work should focus first on external validation. The current evaluation used HAM10000, which was collected under specific dermoscopy protocols. Testing on independent benchmarks such as ISIC 2019, ISIC 2020, BCN20000, and PAD-UFES-20 would pro-

vide stronger evidence about generalisation across acquisition devices, populations, and clinical settings.

Synthetic image quality should also be improved and evaluated more rigorously. Future studies could use Kernel Inception Distance, CLIP-FID, clinical expert scoring, segmentation-guided conditioning, and stronger generative models. Additional experiments should investigate newer backbones such as DINOv2, Swin Transformers, and ConvNeXt variants, as well as calibration analysis for clinical decision support.

Since the synthetic images are generated using a very small subset of real photos, they inherit the limitations of that original data. Any existing biases in lighting, skin tone, or lesion appearance will be copied and potentially magnified by the AI. Therefore, while this method successfully increases the total number of images, it fails to improve the actual diversity of the dataset.

External validation remains the most critical prerequisite for clinical translation. SkinHybrid has been evaluated exclusively on HAM10000, which was acquired under controlled dermoscopy protocols at specific clinical sites, and generalisation to broader populations and acquisition devices is unconfirmed. Future work must validate the model on independent benchmarks including ISIC 2019, ISIC 2020, BCN20000, and PAD-UFES-20.

Additionally, the current SkinHybrid architecture relies on ImageNet-pretrained EfficientNet and ViT initialisations; future iterations should explore stronger and more diverse backbone alternatives, including DINOv2/v3, Swin Transformers, ConvNeXt V2 and their other variants.

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